U-45 CALCULATORS

User Manual

Calculator #30 Sequence & Recursion Grapher (cobweb)

Graphing – colour



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#30 — Sequence & Recursion Grapher (cobweb)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example 

. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

sequence and recursion graphing (time plot, cobweb, phase plot).

Chapters

1. Getting started
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1 · Getting started

Open U-45 CALCULATORS, choose **#30** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#35, #36, #37, #40, #41, #42, #45, #46, #47, #50, #51, #52
op	Operators & execute	#30, #32, #33, #34, #43, #44, #48, #49, #54
fn	Functions	#14, #16, #17, #18, #19, #20, #21, #22, #23, #24, #25, #26, #27, #28, #29, #31, #53
mem	Memory, variables & registers	—
clr	Clear, delete & power	#12, #38, #39

mod	**Shift, modes, menus & navigation**	**#1, #2, #3, #4, #5, #6, #7, #8, #9, #10, #11, #13, #15**

2 · Features

45 features: 31 live on the keys, 9 app/UI features, 5 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE **Three sequence slots u(n), v(n), w(n)**
2. LIVE **Explicit formulas in n or recursive formulas using u(n−1), u(n−2)**
3. LIVE **Cross-references between sequences (e.g. v(n) uses u(n−1))**
4. LIVE **Start values u(nMin), u(nMin+1) for first- and second-order recursions**
5. LIVE **Adjustable nMin / nMax plot range**
6. LIVE **Time plot of terms as glowing dots**
7. LIVE **Cobweb (web) plot with y = x diagonal and staircase iteration**
8. LIVE **Phase plot of u(n) against v(n)**
9. LIVE **Partial-sum Σ display toggle**
10. LIVE **Trace by term index with Δ and ratio read-out**
11. LIVE **Cobweb trace steps through iterations one at a time**
12. LIVE **Trace jump to any n with EXE**
13. LIVE **Tool: single term a(n)**
14. LIVE **Tool: sum of terms between two indices**
15. LIVE **Tool: limit estimate with convergence check**
16. LIVE **Tool: fixed points with |f′| stability test, marked on the cobweb**
17. LIVE **Tool: arithmetic / geometric pattern test**
18. LIVE **Tool: first n reaching a threshold**
19. LIVE **Tool: eventual period (cycle) detection up to 64**
20. LIVE **Zoom fit to terms nMin…nMax**

21. LIVE Plot-mode key cycles TIME → WEB → uv
22. LIVE Table of n with u, v, w columns
23. LIVE Home functions U(n), V(n), W(n)
24. LIVE Home sums sumu(a,b) and sumv(a,b)
25. LIVE Arithmetic progression ap() and apsum()
26. LIVE Geometric progression gp(), gpsum() and gpinf()
27. LIVE Fibonacci fib(n) and Lucas lucas(n)
28. LIVE Memoised evaluation up to 20 000 terms
29. LIVE Re-entrancy guard for mutually recursive definitions
30. LIVE Dark phosphor-green display distinct from function graphers
31. LIVE Window form with 14 fields
32. APP/UI Every key numbered 1…N in reading order — matches the key map in the user manual
33. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a “Classic” toggle
34. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)
35. APP/UI Show / hide key numbers with one button (remembered between sessions)
36. APP/UI Copy the current display text to the clipboard
37. APP/UI Physical-keyboard shortcuts: digits, + – * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i
38. APP/UI Fit / 1:1 zoom that never pushes keys off a phone screen
39. APP/UI Reference photo viewer and tool-tip name on every key

40. APP/UI **Reset = fresh power-on state; works fully offline (PWA cache / Android APK)**
41. PLANNED **Graph image export (PNG)**
42. PLANNED **Shareable function list link**
43. PLANNED **Dark screen theme**
44. PLANNED **Speech read-out of values**
45. PLANNED **Exam mode**

2b · Feature guide

Calculator #30 is a sequence and recursion graphing (time plot, cobweb, phase plot). Each card below is one of its 31 working features, with the keys that reach it and, where one exists, a verified example.

F1 Three sequence slots u(n), v(n), w(n)

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

See it in action: Explicit sequence $w(n) = 3 + 2(-0.8)^n$ at n = 2 — *W(2)*= 4.28

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F2 Explicit formulas in n or recursive formulas using u(n–1), u(n–2)

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

See it in action: Explicit sequence $w(n) = 3 + 2(-0.8)^n$ at n = 2 — *W(2)*= 4.28

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F3 Cross-references between sequences (e.g. v(n) uses u(n−1))

Where: #1 u(n)= Edit sequences; #5 SHIFT SHIFT / 2nd — the next key uses its upper (shift-coloured) function; #34 , Comma / argument separator (between function arguments, list items or coordinates)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F4 Start values u(nMin), u(nMin+1) for first- and second-order recursions

Where: #51 . Decimal point. Pressing it first gives "0."; a second point in the same number is ignored; #2 WINDOW n range, start values and view

See it in action: Second-order recursion v(n) = v(n−1) + v(n−2), v(1) = v(2) = 1 — *V(10)* = 55; Sum of the first 10 terms of 3, 7, 11, … — *apsum(3,4,10)* = 210

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F5 Adjustable nMin / nMax plot range

Where: #1 u(n)= Edit sequences; #2 WINDOW n range, start values and view; #16 WEB Cycle time/web/phase plot

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F6 Time plot of terms as glowing dots

Where: #1 u(n)= Edit sequences; #16 WEB Cycle time/web/phase plot; #18 u(Previous terms of u

See it in action: Sum of the first 10 terms of 3, 7, 11, … — *apsum(3,4,10)*= 210

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F7 Cobweb (web) plot with y = x diagonal and staircase iteration

Where: #1 u(n)= Edit sequences; #16 WEB Cycle time/web/phase plot

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F8 Phase plot of u(n) against v(n)

Where: #1 u(n)= Edit sequences; #16 WEB Cycle time/web/phase plot

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F9 Partial-sum Σ display toggle

Where: #15 MODE SETUP — angle unit, number format, display options; #23 U(Term of u on the home screen; #24 V(Term of v

See it in action: Sum of the first 10 terms of 3, 7, 11, … — *apsum(3,4,10)* = 210; Sum v(1)+…+v(10) of the Fibonacci slot — *sumv(1,10)* = 143

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F10 Trace by term index with Δ and ratio read-out

Where: #6 TRACE TRACE — move along a graph; #17 n Index variable; #21 n−1 Previous index

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F11 Cobweb trace steps through iterations one at a time

Where: #1 u(n)= Edit sequences; #6 TRACE TRACE — move along a graph; #16 WEB Cycle time/web/phase plot

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F12 Trace jump to any n with EXE

Where: #6 TRACE TRACE — move along a graph; #54 ENTER Execute / = — evaluate the line and show the result (EXE / ENTER / =)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F13 Tool: single term a(n)

Where: #23 U(Term of u on the home screen; #24 V(Term of v; #25 W(Term of w

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F14 Tool: sum of terms between two indices

Where: #23 U(Term of u on the home screen; #24 V(Term of v; #18 u(Previous terms of u

See it in action: Sum of the first 10 terms of 3, 7, 11, … — *apsum(3,4,10)* = 210; Sum v(1)+…+v(10) of the Fibonacci slot — *sumv(1,10)* = 143

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F15 Tool: limit estimate with convergence check

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F16 Tool: fixed points with |f′| stability test, marked on the cobweb

Where: #11 TOOLS Sums, limits, fixed points

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F17 Tool: arithmetic / geometric pattern test

Where: #26 AP n-th term of an arithmetic progression; #27 GP n-th term of a geometric progression

See it in action: Infinite geometric series 6 + 3 + 1.5 + … — *gpinf(6,0.5)*= 12

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F18 Tool: first n reaching a threshold

Where: #51 . Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

See it in action: Sum of the first 10 terms of 3, 7, 11, … — *apsum(3,4,10)* = 210

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F19 Tool: eventual period (cycle) detection up to 64

Where: #1 u(n)= Edit sequences; #16 WEB Cycle time/web/phase plot

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F20 Zoom fit to terms nMin…nMax

Where: #3 ZOOM ZOOM — zoom presets; #18 u(Previous terms of u; #19 v(Terms of v

See it in action: Sum of the first 10 terms of 3, 7, 11, … — *apsum(3,4,10)* = 210

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F21 Plot-mode key cycles TIME → WEB → uv

Where: #1 u(n)= Edit sequences; #16 WEB Cycle time/web/phase plot

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F22 Table of n with u, v, w columns

Where: #2 WINDOW n range, start values and view; #4 GRAPH GRAPH — draw the graph

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F23 Home functions U(n), V(n), W(n)

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F24 Home sums sumu(a,b) and sumv(a,b)

Where: #11 TOOLS Sums, limits, fixed points

See it in action: Sum v(1)+…+v(10) of the Fibonacci slot — *sumv(1,10)* = 143

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F25 Arithmetic progression ap() and apsum()

Where: #26 AP n-th term of an arithmetic progression; #27 GP n-th term of a geometric progression

See it in action: Sum of the first 10 terms of 3, 7, 11, … — *apsum(3,4,10)* = 210

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F26 Geometric progression gp(), gpsum() and gpinf()

Where: #27 GP n-th term of a geometric progression; #26 AP n-th term of an arithmetic progression

See it in action: Infinite geometric series 6 + 3 + 1.5 + … — *gpinf(6,0.5)* = 12

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F27 Fibonacci fib(n) and Lucas lucas(n)

Where: #29 Fib Fibonacci number

See it in action: Fibonacci number F(20) — *fib(20)* = 6765; Sum v(1)+…+v(10) of the Fibonacci slot — *sumv(1,10)* = 143

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F28 Memoised evaluation up to 20 000 terms

Where: #18 u(Previous terms of u; #19 v(Terms of v; #20 w(Terms of w

See it in action: Sum of the first 10 terms of 3, 7, 11, … — *apsum(3,4,10)* = 210

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F29 Re-entrancy guard for mutually recursive definitions

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F30 Dark phosphor-green display distinct from function graphers

Where: #5 SHIFT SHIFT / 2nd — the next key uses its upper (shift-coloured) function; #15 MODE SETUP — angle unit, number format, display options; #34 , Comma / argument separator (between function arguments, list items or coordinates)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F31 Window form with 14 fields

Where: #2 WINDOW n range, start values and view

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

3 · Numbered key map



Calculator #30 with key numbers shown (Hi-Contrast colours).

Key table

“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	u(n)=	PLOT▸	·	·	mod
2	WINDOW	TBLSET	·	·	mod
3	ZOOM	·	·	·	mod
4	GRAPH	TABLE	·	·	mod
5	SHIFT	·	·	·	mod
6	TRACE	·	·	·	mod
7	▲	·	·	·	mod
8	▼	·	·	·	mod
9	◀	·	·	·	mod
10	▶	·	·	·	mod
11	TOOLS	·	·	·	mod
12	ON	·	·	·	clr
13	HOME	·	·	·	mod
14	SEL	·	·	·	fn
15	MODE	·	·	·	mod
16	WEB	·	·	·	fn
17	n	·	·	·	fn
18	u(	·	·	·	fn
19	v(	·	·	·	fn
20	w(	·	·	·	fn
21	n−1	·	·	·	fn
22	n−2	·	·	·	fn
23	U(	Σu	·	·	fn

#	Face	Shift	Alpha	3rd	Group
24	V(	Σv	·	·	fn
25	W(	·	·	·	fn
26	AP	APΣ	·	·	fn
27	GP	GPΣ	·	·	fn
28	GP∞	·	·	·	fn
29	Fib	Luc	·	·	fn
30	^	·	·	·	op
31	x^2	√	·	·	fn
32	(	·	·	·	op
33	)	·	·	·	op
34	,	·	·	·	op
35	7	·	·	·	num
36	8	·	·	·	num
37	9	·	·	·	num
38	DEL	·	·	·	clr
39	AC	·	·	·	clr
40	4	·	·	·	num
41	5	·	·	·	num
42	6	·	·	·	num
43	×	·	·	·	op
44	÷	·	·	·	op
45	1	·	·	·	num
46	2	·	·	·	num
47	3	·	·	·	num
48	+	·	·	·	op
49	−	·	·	·	op
50	0	·	·	·	num
51	.	·	·	·	num

#	Face	Shift	Alpha	3rd	Group
52	(–)	Ans	·	·	num
53	π	·	·	·	fn
54	ENTER	·	·	·	op

4 · Key-by-key reference

Every key on calculator #30, in number order, with every function it carries in this app.

#1 u(n)= `Shift, modes, menus & navigation`

Primary: Edit sequences

shifted (2nd / f / SHIFT layer) (press #5 first): PLOT▸ — Cycle time/web/phase plot

#2 WINDOW `Shift, modes, menus & navigation`

Primary: n range, start values and view

shifted (2nd / f / SHIFT layer) (press #5 first): TBLSET — TBLSET — table start and step

#3 ZOOM `Shift, modes, menus & navigation`

Primary: ZOOM — zoom presets

#4 GRAPH `Shift, modes, menus & navigation`

Primary: GRAPH — draw the graph

shifted (2nd / f / SHIFT layer) (press #5 first): TABLE — TABLE — table of values

#5 SHIFT `Shift, modes, menus & navigation`

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#6 TRACE `Shift, modes, menus & navigation`

Primary: TRACE — move along a graph

#7 ▲ `Shift, modes, menus & navigation`

Primary: Cursor ▲ — up / previous history line / previous item

#8 ▼ `Shift, modes, menus & navigation`

Primary: Cursor ▼ — down / next history line / next item

#9 ◀ `Shift, modes, menus & navigation`

Primary: Cursor ◀ — move left in the line

#10 ▶ `Shift, modes, menus & navigation`

Primary: Cursor ▶ — move right in the line

#11 TOOLS `Shift, modes, menus & navigation`

Primary: Sums, limits, fixed points

#12 ON `Clear, delete & power`

Primary: ON — power on

#13 HOME `Shift, modes, menus & navigation`

Primary: HOME — return to the main calculation screen

#14 SEL `Functions`

Primary: sel

#15 MODE `Shift, modes, menus & navigation`

Primary: SETUP — angle unit, number format, display options

#16 WEB `Functions`

Primary: Cycle time/web/phase plot

#17 n `Functions`

Primary: Index variable

#18 u(`Functions`

Primary: Previous terms of u

#19 v(`Functions`

Primary: Terms of v

#20 **w(** Functions

Primary: Terms of w

#21 **n−1** Functions

Primary: Previous index

#22 **n−2** Functions

Primary: Index two back

#23 **U(** Functions

Primary: Term of u on the home screen

shifted (2nd / f / SHIFT layer) (press #5 first): Σu — Sum of u terms

#24 **V(** Functions

Primary: Term of v

shifted (2nd / f / SHIFT layer) (press #5 first): Σv — Sum of v terms

#25 **W(** Functions

Primary: Term of w

#26 AP Functions

Primary: n-th term of an arithmetic progression

shifted (2nd / f / SHIFT layer) (press #5 first): APΣ — Arithmetic series

#27 GP Functions

Primary: n-th term of a geometric progression

shifted (2nd / f / SHIFT layer) (press #5 first): GPΣ — Geometric series

#28 GP∞ Functions

Primary: Sum to infinity

#29 Fib Functions

Primary: Fibonacci number

shifted (2nd / f / SHIFT layer) (press #5 first): Luc — Lucas number

#30 ^ Operators & execute

Primary: Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

Keyboard: [^]

#31 x^2 Functions

Primary: Square x^2

shifted (2nd / f / SHIFT layer) (press #5 first): $\sqrt{}$ — Square root $\sqrt{}$

Keyboard: q

#32 (Operators & execute

Primary: Opening bracket (

Keyboard: (

#33) Operators & execute

Primary: Closing bracket). Closing brackets at the end of a line may be omitted

Keyboard:)

#34 , Operators & execute

Primary: Comma / argument separator (between function arguments, list items or coordinates)

Keyboard: ,

#35 7 Digits & number entry

Primary: Digit 7

Keyboard: 7

#36 8 Digits & number entry

Primary: Digit 8

Keyboard: 8

#37 9 Digits & number entry

Primary: Digit 9

Keyboard: 9

#38 DEL Clear, delete & power

Primary: DEL — delete the character at the cursor

Keyboard: Backspace Delete

#39 AC Clear, delete & power

Primary: AC — clear all of the current calculation

Keyboard: Escape

#40 4 Digits & number entry

Primary: Digit 4

Keyboard: 4

#41 5 Digits & number entry

Primary: Digit 5

Keyboard: 5

#42 **6** `Digits & number entry`

Primary: Digit 6

Keyboard: 6

#43 **×** `Operators & execute`

Primary: Multiplication ×

Keyboard: * x

#44 **÷** `Operators & execute`

Primary: Division ÷

Keyboard: /

#45 **1** `Digits & number entry`

Primary: Digit 1

Keyboard: 1

#46 **2** `Digits & number entry`

Primary: Digit 2

Keyboard: 2

#47 **3** `Digits & number entry`

Primary: Digit 3

Keyboard: 3

#48 **+** `Operators & execute`

Primary: Addition +

Keyboard: `+`

#49 **–** `Operators & execute`

Primary: Subtraction –

Keyboard: `-`

#50 **0** `Digits & number entry`

Primary: Digit 0

Keyboard: `0`

#51 **.** `Digits & number entry`

Primary: Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

Keyboard: `.`

#52 **(–)** `Digits & number entry`

Primary: Negative sign (–) — makes the next number negative. Different from the subtraction key

shifted (2nd / f / SHIFT layer) (press #5 first): Ans — Ans — recall the last answer into the current expression

#53 π Functions

Primary: π (3.14159265…) as an exact constant

Keyboard: p

#54 ENTER Operators & execute

Primary: Execute / = — evaluate the line and show the result (EXE / ENTER / =)

Keyboard: Enter =

5 · Worked examples

7 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Second-order recursion v(n) = v(n−1) + v(n−2), v(1) = v(2) = 1

Calculate **V(10)**. Press **Reset** first.

#24 V(→ #45 1 → #50 0 → #33) → #54 ENTER

Display shows 55 — expected 55. ✔ verified on the app.

Example 5.2 — Logistic-type recursion u(n) = 1.5u − 0.05u² approaches 10

Calculate **U(60)**. Press **Reset** first.

#23 U(→ #42 6 → #50 0 → #33) → #54 ENTER

Display shows 10 — expected 10. ✔ verified on the app.

Example 5.3 — Explicit sequence w(n) = 3 + 2(−0.8)ⁿ at n = 2

Calculate **W(2)**. Press **Reset** first.

#25 W(→ #46 2 → #33) → #54 ENTER

Display shows 4.28 — expected 4.28. ✔ verified on the app.

Example 5.4 — Sum of the first 10 terms of 3, 7, 11, …

Calculate **apsum(3,4,10)**. Press **Reset** first.

#5 SHIFT → #26 APΣ → #47 3 → #34 , → #40 4 → #34 , → #45 1 → #50 0 → #33) → #54 ENTER

Display shows `210` — expected 210. ✔ verified on the app.

Example 5.5 — Infinite geometric series 6 + 3 + 1.5 + …

Calculate **gpinf(6,0.5)**. Press **Reset** first.

#28 GP∞ → #42 6 → #34 , → #50 0 → #51 . → #41 5 → #33) → #54 ENTER

Display shows `12` — expected 12. ✔ verified on the app.

Example 5.6 — Fibonacci number F(20)

Calculate **fib(20)**. Press **Reset** first.

#29 Fib → #46 2 → #50 0 → #33) → #54 ENTER

Display shows `6765` — expected 6765. ✔ verified on the app.

Example 5.7 — Sum v(1)+…+v(10) of the Fibonacci slot

Calculate **sumv(1,10)**. Press **Reset** first.

#5 SHIFT → #24 Σv → #45 1 → #34 , → #45 1 → #50 0 → #33) → #54 ENTER

Display shows 143 — expected 143. ✔ verified on the app.

6 · Tricks & hacks

Hack 1: History reuse

Scroll up in the home screen history and press ENTER/EXE to copy an old entry or answer.

Hack 2: Ans chaining

Start an entry with an operator to continue from the previous answer.

Hack 3: Store variables

Use STO▶ / → to store values into letters A–Z, then use them in expressions.

Hack 4: Table of values

Use the TABLE view to scan where a function changes sign before solving.

Hack 5: Window first

If a graph looks empty, adjust the WINDOW / zoom standard.

Hack 6: Edit instead of retyping

After a result, press ◀ or ▶ to bring the last expression back with the cursor in it; change one number and press = again.

Hack 7: Ans chains

Start a line with an operator (e.g. ×) and the calculator inserts Ans automatically — great for compound growth: 1.05 ×, then = = = ...

Hack 8: Exact vs decimal

Natural-display models keep fractions, √ and π exact. Press S⇔D to flip to decimal; press it again to return.

Hack 9: Close brackets for free

Trailing closing brackets can be left off; = closes them for you.

Hack 10: Implicit multiplication

Type 2π or 3(4+1) without ×. Implicit multiplication binds tighter than ÷ on textbook models — use brackets if in doubt.

Hack 11: Store constants

Store a value you reuse (e.g. g = 9.81) with STO and a letter; then type the letter in formulas.

Hack 12: Power chains

x^y chains right to left: 2^3^2 = 2^9 = 512, not 64. Use brackets: (2^3)^2 = 64.

Hack 13: Angles

Check the D / R / G indicator before trig. sin 30 = 0.5 only in degrees.

Hack 14: Memory sums

M+ / M− keep a running total separate from the calculation line; RCL M shows it.

Hack 15: Replay history

Use ▲ ▼ to scroll earlier lines; press = on an old line to re-run it.

Hack 16: Hidden functions on this model

10 extra functions sit on shifted layers. Press the layer key once, then the key: for example #5 then #1 → PLOT▶; #5 then #2 → TBLSET; #5 then #4 → TABLE; #5 then #23 → ∑u.

Hack 17: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left→right, top→bottom from #1 to #54.

Hack 18: Type faster on a computer

24 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 19: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 20: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 21: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
^	#30	^	Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)
q	#31	x^2	Square x^2
(	#32	(	Opening bracket (
)	#33	)	Closing bracket). Closing brackets at the end of a line may be omitted
,	#34	,	Comma / argument separator (between function arguments, list items or coordinates)
7	#35	7	Digit 7
8	#36	8	Digit 8
9	#37	9	Digit 9
Backspace Delete	#38	DEL	DEL — delete the character at the cursor
Escape	#39	AC	AC — clear all of the current calculation
4	#40	4	Digit 4
5	#41	5	Digit 5
6	#42	6	Digit 6
* x	#43	×	Multiplication ×
/	#44	÷	Division ÷
1	#45	1	Digit 1
2	#46	2	Digit 2
3	#47	3	Digit 3
+	#48	+	Addition +
-	#49	−	Subtraction −

0	#50	0	Digit 0
.	#51	.	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored
p	#53	π	π (3.14159265…) as an exact constant
Enter =	#54	ENTER	Execute / = — evaluate the line and show the result (EXE / ENTER / =)

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: (–) vs –

Use the negative key for negative numbers.

Mistake 2: Radian mode

Graphs of sin x in degrees look flat over −10..10; switch to radians for standard trig windows.

Mistake 3: Implicit multiplication

2/3x may be read as 2/(3x) depending on the model; use brackets.

Mistake 4: Using – instead of (–)

The negative key (–) and the subtraction key are different; "– 5" at the start uses Ans – 5.

Mistake 5: Wrong angle unit

If sin 30 gives −0.988, the calculator is in radians. Switch to degrees in SETUP.

Mistake 6: -3^2 vs $(-3)^2$

$-3^2 = -9$ because the square is applied first. Type $(-3)^2$ to get 9.

Mistake 7: Division precedence

1 ÷ 2π may be read as 1 ÷ (2π) on textbook models. Type 1 ÷ (2 × π) explicitly to be sure.

Mistake 8: Forgetting SHIFT is one-shot

SHIFT applies only to the next key. If the S indicator is still lit, press SHIFT again to cancel.

Mistake 9: Clearing memory by accident

AC clears the line but not memory; CLR (SHIFT 9) can clear memory and variables.

Mistake 10: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 11: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

Mistake 12: Minus vs negative

The (−) negative key and the − subtraction key are different. Use (−) to start a negative number, − between two numbers.

9 · Practice drills

Set A — Calculation drills (7)

Work each one on calculator #30, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. **V(10)** = _____
2. **U(60)** = _____
3. **W(2)** = _____
4. **apsum(3,4,10)** = _____
5. **gpinf(6,0.5)** = _____
6. **fib(20)** = _____
7. **sumv(1,10)** = _____

Set B — Key-finding drills (54)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. **Which key is "u(n)=" (Edit sequences)?** _____
2. **Which key is "WINDOW" (n range, start values and view)?** _____
3. **Which key is "ZOOM" (ZOOM)?** _____
4. **Which key is "GRAPH" (GRAPH)?** _____
5. **Which key is "SHIFT" (SHIFT / 2nd)?** _____
6. **Which key is "TRACE" (TRACE)?** _____
7. **Which key is " ▲ " (Cursor ▲)?** _____

8. Which key is “▼” (Cursor ▼)? _____

9. Which key is “◀” (Cursor ◀)? _____

10. Which key is “▶” (Cursor ▶)? _____

11. Which key is “TOOLS” (Sums, limits, fixed points)? _____

12. Which key is “ON” (ON)? _____

13. Which key is “HOME” (HOME)? _____

14. Which key is “SEL” (sel)? _____

15. Which key is “MODE” (SETUP)? _____

16. Which key is “WEB” (Cycle time/web/phase plot)? _____

17. Which key is “n” (Index variable)? _____

18. Which key is “u(” (Previous terms of u)? _____

19. Which key is “v(” (Terms of v)? _____

20. Which key is “w(” (Terms of w)? _____

21. Which key is “n−1” (Previous index)? _____

22. Which key is “n−2” (Index two back)? _____

23. Which key is “U(” (Term of u on the home screen)? _____

24. Which key is “V(” (Term of v)? _____

25. Which key is “W(” (Term of w)? _____

26. Which key is “AP” (n-th term of an arithmetic progression)? _____

27. Which key is “GP” (n-th term of a geometric progression)? _____

28. Which key is “GP∞” (Sum to infinity)? _____

29. Which key is “Fib” (Fibonacci number)? _____

30. Which key is “^” (Power x^y / ^)? _____

31. Which key is “x^2” (Square x^2)? _____

32. Which key is “(” (Opening bracket)? _____

33. Which key is “)” (Closing bracket). Closing brackets at the end of a line may be omitted)? _____

34. Which key is “,” (Comma / argument separator)? _____

35. Which key is “DEL” (DEL)? _____

36. Which key is “AC” (AC)? _____

37. Which key is “×” (Multiplication ×)? _____

38. Which key is “÷” (Division ÷)? _____

39. Which key is “+” (Addition +)? _____

40. Which key is “−” (Subtraction −)? _____

41. Which key is “.” (Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored)? _____

42. Which key is “(−)” (Negative sign)? _____

43. Which key is “π” (π)? _____

44. Which key is “ENTER” (Execute / =)? _____

45. How do you reach “PLOT▶”? _____

46. How do you reach “TBLSET”? _____

47. How do you reach “TABLE”? _____

48. How do you reach “Σu”? _____

49. How do you reach “Σv”? _____

50. How do you reach “APΣ”? _____

51. How do you reach “GPΣ”? _____

52. How do you reach “Luc”? _____

53. How do you reach “√”? _____

54. How do you reach “Ans”? _____

Appendix A — Drill answers

Set A

1. **55** (display `55`)
   `#24` V(→ `#45` 1 → `#50` 0 → `#33`) → `#54` ENTER
2. **10** (display `10`)
   `#23` U(→ `#42` 6 → `#50` 0 → `#33`) → `#54` ENTER
3. **4.28** (display `4.28`)
   `#25` W(→ `#46` 2 → `#33`) → `#54` ENTER
4. **210** (display `210`)
   `#5` SHIFT → `#26` APΣ → `#47` 3 → `#34` , → `#40` 4 → `#34` , → `#45` 1 → `#50` 0 → `#33`) → `#54` ENTER
5. **12** (display `12`)
   `#28` GP∞ → `#42` 6 → `#34` , → `#50` 0 → `#51` . → `#41` 5 → `#33`) → `#54` ENTER
6. **6765** (display `6765`)
   `#29` Fib → `#46` 2 → `#50` 0 → `#33`) → `#54` ENTER
7. **143** (display `143`)
   `#5` SHIFT → `#24` Σv → `#45` 1 → `#34` , → `#45` 1 → `#50` 0 → `#33`) → `#54` ENTER

Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14
15. #15
16. #16
17. #17
18. #18
19. #19
20. #20
21. #21
22. #22
23. #23
24. #24
25. #25
26. #26
27. #27
28. #28
29. #29
30. #30
31. #31
32. #32
33. #33
34. #34
35. #38
36. #39
37. #43
38. #44
39. #48
40. #49
41. #51
42. #52
43. #53
44. #54
45. #5 SHIFT then #1 u(n)=
46. #5 SHIFT then #2 WINDOW
47. #5 SHIFT then #4 GRAPH
48. #5 SHIFT then #23 U(
49. #5 SHIFT then #24 V(
50. #5 SHIFT then #26 AP
51. #5 SHIFT then #27 GP
52. #5 SHIFT then #29 Fib
53. #5 SHIFT then #31 x^2
54. #5 SHIFT then #52 (-)

Appendix B — Function finder (A–Z)

Function	Label	Keys
AC — clear all of the current calculation	AC	#39
Addition +	+	#48
Ans — recall the last answer into the current expression	Ans	#5 → #52
Arithmetic series	APΣ	#5 → #26
Closing bracket). Closing brackets at the end of a line may be omitted	)	#33
Comma / argument separator (between function arguments, list items or coordinates)	,	#34
Cursor ▲ — up / previous history line / previous item	▲	#7
Cursor ▶ — move right in the line	▶	#10
Cursor ▼ — down / next history line / next item	▼	#8
Cursor ◀ — move left in the line	◀	#9
Cycle time/web/phase plot	PLOT▸	#5 → #1
Cycle time/web/phase plot	WEB	#16
Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored	.	#51
DEL — delete the character at the cursor	DEL	#38
Digit 0	0	#50
Digit 1	1	#45
Digit 2	2	#46
Digit 3	3	#47
Digit 4	4	#40
Digit 5	5	#41
Digit 6	6	#42
Digit 7	7	#35
Digit 8	8	#36

Digit 9	9	#37
Division ÷	÷	#44
Edit sequences	u(n)=	#1
Execute / = — evaluate the line and show the result (EXE / ENTER / =)	ENTER	#54
Fibonacci number	Fib	#29
Geometric series	GPΣ	#5 → #27
GRAPH — draw the graph	GRAPH	#4
HOME — return to the main calculation screen	HOME	#13
Index two back	n−2	#22
Index variable	n	#17
Lucas number	Luc	#5 → #29
Multiplication ×	×	#43
n range, start values and view	WINDOW	#2
n-th term of a geometric progression	GP	#27
n-th term of an arithmetic progression	AP	#26
Negative sign (−) — makes the next number negative. Different from the subtraction key	(−)	#52
ON — power on	ON	#12
Opening bracket (	(	#32
Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)	^	#30
Previous index	n−1	#21
Previous terms of u	u(	#18
sel	SEL	#14
SETUP — angle unit, number format, display options	MODE	#15
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	SHIFT	#5
Square root √	√	#5 → #31
Square x^2	x^2	#31

Subtraction –	–	#49
Sum of u terms	Σu	#5 → #23
Sum of v terms	Σv	#5 → #24
Sum to infinity	GP∞	#28
Sums, limits, fixed points	TOOLS	#11
TABLE — table of values	TABLE	#5 → #4
TBLSET — table start and step	TBLSET	#5 → #2
Term of u on the home screen	U(	#23
Term of v	V(	#24
Term of w	W(	#25
Terms of v	v(	#19
Terms of w	w(	#20
TRACE — move along a graph	TRACE	#6
ZOOM — zoom presets	ZOOM	#3
π (3.14159265…) as an exact constant	π	#53

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

Sequence & recursion grapher. In a recursive slot, u(n−1) means the previous term; start values are set in WINDOW.

The cobweb plot substitutes x for the previous term to draw y = f(x) and iterates from u(nMin).

Planned (not yet working)

- **Graph image export (PNG)**
- **Shareable function list link**
- **Dark screen theme**
- **Speech read-out of values**
- **Exam mode**

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 u(n)= ⇧ PLOT▸	#2 WINDOW ⇧ TBLSET	#3 ZOOM
#4 GRAPH ⇧ TABLE	#5 SHIFT	#6 TRACE
#7 ▲	#8 ▼	#9 ◀
#10 ▶	#11 TOOLS	#12 ON
#13 HOME	#14 SEL	#15 MODE
#16 WEB	#17 n	#18 u(
#19 v(	#20 w(	#21 n−1
#22 n−2	#23 U(⇧ Σu	#24 V(⇧ Σv
#25 W(	#26 AP ⇧ APΣ	#27 GP ⇧ GPΣ
#28 GP∞	#29 Fib ⇧ Luc	#30 ^
#31 x^2 ⇧ √	#32 (	#33)
#34 ,	#35 7	#36 8
#37 9	#38 DEL	#39 AC
#40 4	#41 5	#42 6
#43 ×	#44 ÷	#45 1
#46 2	#47 3	#48 +
#49 −	#50 0	#51 .

#52 (–) ⇧ Ans	**#53 π**	**#54 ENTER**

U-45 CALCULATORS · Calculator #30 · cut out and keep